SCIENCE PAGE:

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The body is built to regulate and repair itself. However, when environmental toxins accumulate in the bloodstream, they can place sustained demand on the immune system. Over time, that can drive persistent inflammation, leading to chronic diseases in the future. Understanding what circulates in the blood is the first step in identifying sources of inflammatory strain and reducing long-term disease risk.

Metals

Toxic elements that accumulate in blood and tissues and disrupt cellular function.

Exposure to certain heavy metals can occur through water, food, air, or occupational environments. Once in the body, these elements may circulate in the bloodstream and deposit in tissues.

Unlike many compounds, metals do not readily break down. Over time, accumulation may interfere with enzyme function, increase oxidative stress, impair mitochondrial energy production, and alter immune signaling.

Heavy Metals

Lead

Lead exposure has been associated with neurologic dysfunction, cardiovascular strain, kidney impairment and developmental effects in children. Even low levels may contribute to long-term systemic burden.

Arsenic

Arsenic is often encountered through contaminated water or certain foods and has been linked to vascular disease, metabolic dysfunction, immune imbalance and increased cancer risk with chronic exposure.

When metal burden persists, it can create cumulative stress across organ systems. For appropriate patients, identifying and reducing that burden may be an important step toward restoring physiologic balance.

Agricultural Chemicals

Chemicals used in food production that can affect metabolic and neurologic pathways.

Modern agriculture relies on chemical agents to protect crops and increase yield. While regulatory standards aim to limit exposure, trace amounts of certain compounds can enter the food and water supply.

Some of these chemicals are biologically active. When present in the body, they may interact with metabolic enzymes, hormone signaling pathways and neurologic processes. Depending on the compound, residues can persist in tissues over time.

PESTICIDES
Glyphosate

Glyphosate is one of the most widely used herbicides in the world. While much of it is eliminated from the body relatively quickly, repeated exposure may contribute to cumulative burden.

Research has explored how glyphosate may influence gut microbiome balance, increase oxidative stress, and interfere with certain cellular pathways involved in metabolism and immune regulation. Disruption in these systems has been associated with chronic inflammation, metabolic dysfunction, and broader long-term health concerns.

Legacy Insecticides
p,p′-DDE (DDT)

p,p′-DDE is a breakdown product of DDT, a pesticide that was banned decades ago but still lingers in soil, water, and the food supply because it breaks down very slowly.

This compound is stored in body fat, which means it can remain in the body for years after exposure. Over time, studies have linked higher levels to disruptions in hormone balance and increased risk of certain long-term health conditions.

Volatile & Industrial Chemicals

Airborne and solvent-based chemicals commonly found in indoor and occupational environments.

Many everyday materials, including paints, cleaning products, fuels, adhesives and building materials release airborne chemicals. These compounds can be inhaled in small amounts over time, particularly in enclosed or poorly ventilated spaces.

Some of these chemicals are rapidly processed by the body. Others, especially with repeated exposure, may contribute to cumulative strain on the nervous system, liver, or blood-forming systems.

Volatile Organic Compounds (VOCs)
Toluene

Toluene is a solvent found in products such as paint thinners, adhesives and gasoline. Short-term exposure at high levels can affect the nervous system. With ongoing exposure, research has explored potential impacts on cognitive function, liver health, and cellular stress pathways.

Aromatic Hydrocarbons
Benzene

Benzene is a widely studied industrial chemical present in fuels, tobacco smoke and some manufacturing environments. It is known to affect the bone marrow, where blood cells are produced. Chronic exposure has been associated with increased risk of certain blood disorders and cancers.

Persistent Organic Pollutants (POPs)

Long-lasting synthetic chemicals that can accumulate in the body over time.

Some industrial and consumer chemicals are designed to resist heat, water and degradation. That durability makes them useful in manufacturing, but it also means they break down slowly in the environment and in the human body.

Because many of these compounds are fat-soluble, they can accumulate in tissues and remain present for years. Over time, ongoing exposure may contribute to immune disruption and increased inflammatory stress.

PFAS / Fluorinated Compounds

PFOS

PFAS are a large group of chemicals used in non-stick cookware, stain-resistant fabrics, firefighting foams and food packaging. PFOS is one of the most studied compounds in this class. Research has linked long-term exposure to changes in immune response, cholesterol levels, thyroid function, and increased risk of certain chronic diseases.

PCBs

PCB-153

Polychlorinated biphenyls (PCBs) were widely used in electrical equipment and industrial materials before being banned in many countries. PCB-153 is a commonly detected variant. Due to their persistence, PCBs remain present in soil, water, and food chains. Chronic exposure has been associated with metabolic dysfunction, endocrine disruption, and increased cancer risk.

Flame Retardants

PBDE-47

PBDEs are flame-retardant chemicals used in furniture, electronics, and textiles. PBDE-47 is one of the most frequently detected forms in human tissue. Studies have explored potential links to hormone disruption, neurodevelopmental effects, and thyroid imbalance.

Endocrine Disrupting Chemicals (EDCs)

Chemicals that can interfere with the body's hormone signaling, even at low levels of exposure.

Hormones are the body's internal messengers. They regulate metabolism, reproduction, mood, growth and immune balance. Certain synthetic compounds are structurally similar to natural hormones. When present in the body, they may mimic, block, or alter normal signaling patterns. Over time, this interference can contribute to metabolic imbalance, reproductive changes, and chronic inflammatory stress.

Phenols

Bisphenol A (BPA)

Commonly found in plastics, food containers and can linings, BPA has been studied for its ability to act like estrogen in the body. Higher cumulative exposure has been associated with shifts in metabolic health, cardiovascular markers and reproductive function.

Phthalates

MEHP

Phthalates are used to soften plastics and stabilize fragrances in personal care products. MEHP is a marker of phthalate exposure in the body. Research suggests links between sustained exposure and altered hormone levels, fertility concerns and metabolic disruption.

Parabens

Methylparaben

Parabens are preservatives used in cosmetics and pharmaceuticals. Methylparaben is among the most commonly detected forms in human tissue. Studies have explored potential effects on estrogen signaling and long-term endocrine balance.

Biologic Toxins

Naturally occurring compounds produced by certain organisms that can enter the food chain.

Not all biologic burden comes from synthetic chemicals. Some toxins are produced naturally by molds and fungi, particularly in conditions of moisture or improper storage. These compounds are biologically active. Depending on dose and duration, they may influence immune regulation, oxidative stress pathways and organ function.

Mycotoxins

Ochratoxin A. Aflatoxin M1.

Mycotoxins are toxic substances produced by specific species of mold. Ochratoxin A has been studied for its potential effects on kidney health and immune signaling. Aflatoxin M1, a metabolite that can appear in dairy products when livestock consume contaminated feed, has been associated with liver stress and long-term cancer risk at sustained levels of exposure.

Because these toxins can persist in the bloodstream, even low-level exposure over time may contribute to inflammatory strain. For some patients, understanding biologic toxin burden is part of restoring clearer immune balance and physiologic stability.

Microplastics

Microscopic plastic particles that can circulate in the bloodstream and tissues.

As larger plastic materials break down, they form particles small enough to enter air, water and food. In recent years, microplastics have been detected in human blood and other tissues, reflecting the scale of environmental exposure.

Because of their size, some particles may move beyond the digestive tract and into circulation. Early research suggests they may contribute to inflammatory signaling or act as carriers for other environmental chemicals. Scientific understanding is still evolving, particularly regarding long-term health implications.

Plastic Polymers

Polyethylene Fragments

Polyethylene is one of the most widely produced plastics worldwide. As products degrade, microscopic fragments can form and enter the environment. These particles have been identified in human biological samples, raising questions about cumulative exposure over time.

While definitive conclusions are still being studied, assessing microplastic presence may help provide a clearer picture of total environmental burden and its potential role in chronic inflammatory stress.

Biologic Persistence

Pathogens or pathogen-derived materials that may remain in the body and contribute to prolonged immune activation.

In some individuals, the effects of an infection may extend beyond the initial illness. Even after acute symptoms improve, traces of the organism, or the immune response it triggered, can continue to influence the body. When immune activation remains elevated instead of returning to baseline, it may contribute to ongoing inflammation and physiologic strain.

Bacterial Pathogens

Borrelia burgdorferi (Lyme disease)

Borrelia burgdorferi is the bacterium associated with Lyme disease. While many patients recover fully with appropriate treatment, some experience ongoing symptoms that may involve persistent immune activation or inflammatory signaling.

Persistent Pathogen-Associated Proteins

Spike Protein

Following viral exposure or vaccination, the body generates viral proteins as part of its immune response. In certain contexts, fragments of these proteins may remain detectable for a period of time. Ongoing research is examining how long such materials persist and whether they play a role in prolonged immune activation in select individuals.

Chronic Viral Reactivation

EBV. CMV.

Some viruses, including Epstein–Barr virus (EBV) and cytomegalovirus (CMV), remain dormant in the body after initial infection. Under conditions of stress or immune imbalance, they may reactivate. Persistent or repeated reactivation has been associated with chronic fatigue, immune dysregulation and inflammatory strain in some patients.

TREATMENT PAGE:

Header copy on the page stays the same.

Would like to add a drop down FAQ below it
(user sees the question, clicks to expand to see the answer type thing).

FAQ:

What is Inuspheresis™?

↓

Inuspheresis™ is an advanced form of Double Filtration Plasma Pheresis (DFPP) that has been used in more than 100,000 treatments for over a decade.

It uses a proprietary second-stage plasma filter engineered to remove circulating environmental toxicants and large inflammatory mediators while preserving beneficial plasma components such as albumin.

Unlike traditional therapeutic plasma exchange (TPE), Inuspheresis™:

- Does not require donor plasma or albumin
- Returns the patient's own filtered plasma

Unlike highly selective adsorption systems:

- It removes a broader range of molecules based on size

What does Inuspheresis™ remove?

↓

Inuspheresis™ is designed to reduce circulating environmental toxicants and large inflammatory carrier molecules from the bloodstream.

Substances shown to be reduced in circulation include:

Environmental toxicants

- Heavy metals (e.g., lead, mercury)
- Agricultural chemicals (e.g., glyphosate, DDE)
- Volatile organic compounds (e.g., benzene)
- Persistent organic pollutants (e.g., PFAS, PCBs, flame retardants)
- Consumer and industrial chemicals (e.g., BPA, phthalates, parabens)
- Biological toxins (e.g., ochratoxin A)
- Microplastic fragments (e.g., polyethylene polymers)

Biologic persistence markers

- Pathogen-associated proteins
- Borrelia-associated markers

Carrier and inflammatory molecules

- Lipoproteins (LDL, Lp(a))
- Immunoglobulins (IgG, IgM)
- Inflammatory proteins (CRP, TNF-α, IL-1β, IL-6, IL-18, IFN-γ)
- Clotting factors (Fibrinogen, Prothrombin, Factor VIII, Factor X)

On average, approximately 50% reduction of lipoproteins and immunoglobulins is observed per treatment session.

Because many environmental toxicants are lipophilic and protein-bound, reducing these carrier molecules can significantly decrease circulating toxic burden.

Ongoing research is focused on understanding long-term changes in total body burden.

How is Inuspheresis™ different from other blood filtration procedures?

↓

Inuspheresis™ differs from traditional plasma exchange in that it:

- Does not require donor plasma
- Returns the patient's own filtered plasma
- Is engineered specifically to reduce environmental and inflammatory burden

Its second-stage membrane is designed to remove larger molecular structures while preserving essential plasma components.

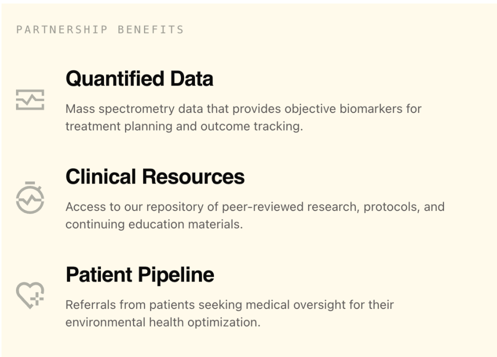
When will Inuspheresis™ be available in the United States?

Inuspheresis™ is currently undergoing the FDA approval process.

Proxima Health anticipates U.S. availability in 2027, pending regulatory clearance.

PRACTITIONERS PAGE:

Remove this section:



Replace quote with this one from Carlos:

"Much of what drives chronic disease has lived in the background — unseen, unmeasured, and untreated. Environmental toxins are one of those forces. Proxima was created to bring them into focus, and to give people the chance to act before illness becomes inevitable"

Insert FAQ section:

What is Proxima Health?

Proxima Health is focused on reducing environmental and biologic burden through advanced blood filtration technology and diagnostics.

Our mission is to help patients live with less inflammatory strain and greater physiologic resilience, grounded in science, transparency and regulatory rigor.

We hold the exclusive U.S. license to bring Inuspheresis™ through FDA approval and into clinical practice.

How is Proxima related to INUS / Ayus?

Inuspheresis™ technology is developed by INUS in Europe.

Proxima Health holds a 20-year exclusive U.S. license to:

- Lead FDA approval
- Commercialize the system
- Provide practitioner training and support

We will serve as the exclusive U.S. distributor.

Will there be territorial exclusivity?

We will not offer formal geographic exclusivity.

However, our economic model is aligned with practitioner success. A device can perform approximately 500 treatments per year at full utilization. We do not intend to oversaturate markets. Expansion will be based on utilization and clinical demand.

What does signing an LOI mean?

An LOI is not a binding contract and does not require a deposit. Contracts cannot be executed prior to FDA clearance.

The LOI signals mutual intent and places your clinic on the waitlist. Participants receive regulatory updates and early access to diagnostics and future products.

When will Inuspheresis™ be available in the U.S.?

FDA submission is underway. We anticipate U.S. clearance in 2027, pending approval.

We maintain an active dialogue with the FDA and are advised by experienced former regulatory leadership.

Is early access possible prior to FDA clearance?

No.

Proxima is committed to full regulatory compliance. This technology is being developed as a long-term, evidence-driven platform and will not bypass U.S. regulatory standards.

What is the anticipated pricing?

Preliminary projections (subject to change):

- Device: $50,000–$60,000
- Single-use filter kit: $2,000–$3,000 per treatment

Clinics typically project patient pricing in the $5,000–$6,000 range, consistent with lower-end TPE pricing in the U.S.

Long-term, our objective is cost reduction through scale and manufacturing efficiencies to improve accessibility.

Will financing or leasing be available?

Initial rollout will likely prioritize direct purchase. Financing and leasing options are expected to follow as adoption expands.

How does the treatment work?

Blood is accessed peripherally and separated into plasma and cells.

The plasma passes through a second-stage membrane engineered to remove molecules larger than albumin. Filtered plasma is then returned to the patient. No donor plasma or albumin is required.

Essential components such as albumin, hormones, minerals and growth factors are preserved.

What does it remove?

On average, a single session reduces:

- ~50% of lipoproteins and immunoglobulins
- ~30–75% reduction of certain circulating toxicants (toxin-dependent)

Because many environmental toxicants are lipophilic and protein-bound, reducing carrier molecules significantly decreases circulating burden.

Ongoing research is evaluating longer-term tissue effects.

What is the standard protocol?

Most patients receive:

- Day 1: Treatment
- Day 2: Rest
- Day 3: Treatment

More complex cases may require three sessions (Days 1, 3, and 5).

Annual maintenance protocols are under evaluation.

How long does treatment take?

Approximately 90–120 minutes, depending on plasma volume and access flow rate.

How safe is it?

In European use (>100,000 treatments), reported adverse events are <1%, typically mild and access-related.

No donor plasma is used, which reduces risks associated with replacement fluids seen in TPE.

Appropriate nursing experience (ER, ICU, nephrology) is strongly recommended.

How does this compare to TPE, EBOO, chelation, or LDL apheresis?

TPE discards plasma and replaces it with donor fluid. Inuspheresis™ filters and returns the patient’s own plasma.

EBOO exposes blood to ozone; it does not filter toxins and follows a different risk and regulatory profile.

Chelation primarily targets heavy metals.

LDL apheresis selectively removes LDL.

Inuspheresis™ removes molecules larger than albumin (~10 nm), capturing a broader range of carrier-bound toxicants.

How many staff are required?

Typically one nurse per patient. One nurse may manage multiple patients depending on clinic workflow.

What training is provided?

A five-day training program (centralized or onsite) covering:

- Apheresis fundamentals
- Device operation
- Supervised treatments

Certification is required. A licensed physician must be onsite. Certified nursing staff must operate the system.

What are clinic prerequisites?

At minimum:

- Licensed physician onsite
- Qualified nursing staff

Additional site criteria will be shared prior to launch.

Is this covered by insurance?

Initially, this will be cash-pay.

Longer-term, we plan to pursue reimbursement pathways by expanding existing CPT frameworks and generating the required clinical evidence.

Will you offer diagnostic testing?

Yes.

Pre- and post-treatment blood testing, as well as eluate analysis is planned.

Diagnostics will:

- Support patient education
- Strengthen clinical validation
- Generate publishable data
- Guide personalized protocols

Why are you transparent about uncertainty?

Because credibility requires it.

We clearly distinguish between:

- What is established
- What is under investigation
- What requires further study

This approach builds durable trust and ensures Proxima remains science-first and evidence-driven.

ABOUT PAGE:

Updated bio for Carlos:

Carlos Schuster is Co-Founder and CEO of Proxima Health. His work is rooted in a personal experience: after his sister was diagnosed with cancer at a young age, clinicians suspected early toxic exposure may have played a role. That question, whether harmful exposures could be identified and reduced before disease develops, shaped his path. Carlos earned degrees in materials science from Imperial College London and a PhD in tissue engineering from the University of Cambridge before building and scaling teams in early-stage growth environments. He founded Proxima Health to focus on preventing chronic disease by measuring and reducing environmental burden at its source.

DIAGNOSTICS PAGE:

Proxima Health

Baseline™

A clinically validated at-home test designed to measure environmental toxin load with precision. Built on proven science, so you can understand what's in your blood and move forward informed.

✓ MASS SPECTROMETRY ✓ CLIA CERTIFIED

✓ 150+ BIOMARKERS

Remove Mass Spectrometry. Change 150+ biomarkers to ‘18 Biomarkers tested’

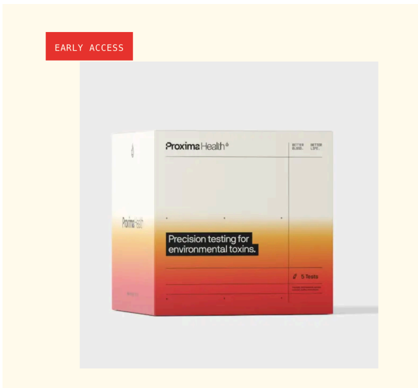
Section below next to test kit:

Every reset begins with the truth

Proxima Health Baseline™ provides an honest assessment of the environmental toxins currently in your bloodstream. By quantifying your exposure to critical toxin markers, we establish the scientific foundation required to begin your journey toward recovery and longevity.

What we test for:
- heavy metals
- PFAS
- microplastics
- pesticides
- BPA
- endocrine disruptors
- mold

- solvents



Remove early access

For the what we measure section:

Heavy Metals
Lead
Mercury

Agricultural Chemicals
Glysophate
p,p’-DDE

Solvents
Benzene

Persistent Organic Pollutants (POPs)
PFAS
PFOA
PCB-126
PBDE-47 (Flame Retardant)

Industrial Chemicals
Phenol - BPA
Phthalates - MEHP
Parabens - Methylparaben

Biological Toxins
Mycotoxins (mold) - Ochratoxin A

Pathogen-associated proteins
Spike Proteins

ADD ONS

Microplastics
Polyethylene

Bacterial Pathogens
Lyme Disease

Remove how it works section

Section next to testing kit with price and everything to still be updated based on information from client.